

Enhanced Ecosystem Monitoring in New York City's Only Old Growth Forest

Abstract: As the world's first old growth, urban research forest the Thain Family Forest (TFF) at the New York Botanical Garden (NYBG) can provide an invaluable reference for regional forest ecosystem monitoring with upgraded data to address emerging needs. Climate change mitigation and adaptation require healthy forests, especially in cities. However, urban forest monitoring lacks an old-growth reference for soil carbon (C). Furthermore, urban soils may pose unique risks with increasingly heavy rainfall, which can remobilize accumulated heavy metals. This project will upgrade the TFF monitoring program to better understand current threats, historical trends and future directions of the urban forested landscape in the Northeast through four related objectives. First, we will follow an international standard for soil C monitoring aligned with the decades-old TFF Continuous Forest Inventory (CFI) program. Second, we will leverage existing hydrological instrumentation and the Soil Climate Analysis Network protocol by deploying the first local weather station focused on hydrology. Third, we will restart the first regional study of forest urbanization by remeasuring soil health, including heavy metal concentrations. Finally, we will implement our project in a sustainable, inclusive and equitable way by supporting the NYBG internship program and coordinating with a regional urban forest advocacy group.

Rationale: The Thain Family Forest, a 20 ha stand within NYBG, Bronx, NY, is the only old growth forest in New York City and the epicenter for 125 years of urban forest inventory and ecosystem monitoring (Britton 1906, McDonnell et al. 1997, Nagele et al. 2024, Fig. 1). With its first flora completed in 1898, the TFF provides the longest record of forest dynamics in North America's densest city (Britton 1906). Complementing floristics, ecosystem monitoring began in the 1980s with the Institute for Ecosystem Studies Urban-Rural Gradient project, which determined that the TFF's primeval ambiance belied uniquely urban soil impairment from heavy metal contamination. (McDonnell et al. 1997, Pouyat and McDonnell, 1991). The tension between the unique land-use history and urbanized context was apparent in the most recent synthesis of the five-year recurring CFI program. Despite intense compositional change driven by pests, pathogens, invasive species and extreme weather, overstory structure has not significantly changed since 1938, while native richness has significantly increased (Nagele 2024).

While aboveground monitoring testifies to the forest's resilience, considerably less is known about how aboveground dynamics connect to belowground C, its sensitivity to extreme weather and implications for pollution mobilization. Given the strong interest in urban forest ecosystem services (Pregitzer et al. 2022) and reforestation efforts (Ward et al, 2021, Oberle et al. 2023), upgraded monitoring of TFF soils will establish an invaluable new reference for Northeastern forests.

This project will upgrade long-term monitoring the Thain Family Forest in alignment with regional standards to inform the intersection between forest carbon cycling, hydrology and soil health.

Objective 1: Soil C monitoring aligned with aboveground inventory and international standards

To understand the relationship between forest inventory, carbon and management, this project will align the CFI plots with an international standard for monitoring soil C stocks. By following the Smithsonian Institution Forest Global Earth Observatory (ForestGEO) protocol, which is also used at Harvard Forest, our inventory data will generate the first high resolution map of soil bulk density and C in an old growth urban forest as a regional reference for urban soil C storage and dynamics.

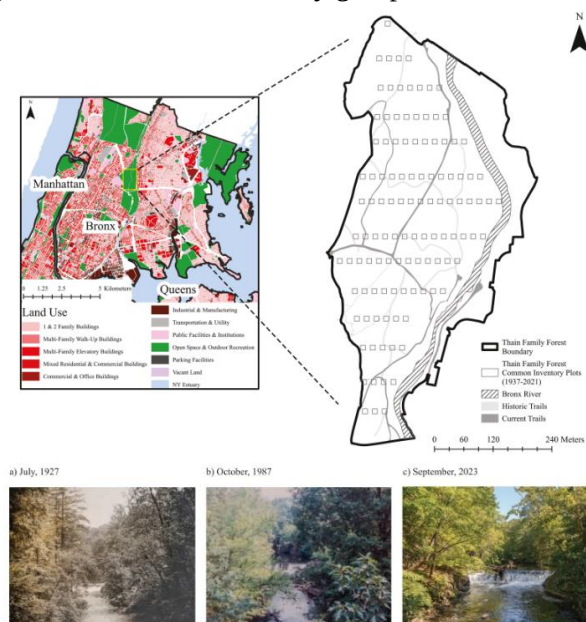


Figure 1: Landover map of the Bronx, NY with the Thain Family Forest, including the Continuous Forest Monitoring plot locations (top, Nagele et al 2024) and repeated imagery illustrating compositional change spanning 100 years at the Bronx Dam (Bottom, Nagele et al. 2023).

To establish a continuous record of forest-water relations, this project will begin monitoring rainfall, humidity, temperature and soil moisture. By aligning with the Soil Climate Analysis Network protocol, the project will inform regional comparisons by representing an extremely urbanized soil and complement existing hydrological monitoring in the TFF, including sap flux sensors and stream gauge.

To establish how climate change and management impact urban forest soil health, we will rescue a historic dataset of heavy metal contamination and complement it with long term soil health monitoring that aligns with regional standards. Doing so will demonstrate how legacy pollution changes and inform management activities, including the planned installation of a fish passage project on the Bronx River.

To ensure long-term sustainability and inclusivity, the project will align with institutional strategic objectives and complement budgeted programs in the TFF, the Bronx River Watershed and Urban Conservation. We will engage youth in data collection through the NYBG Internship Program. All data will be made publicly available through a new databasing effort for the TFF.

Objective 1: To quantify C following ForestGEO protocols, we will split samples for parallel analyses. The first sample of approximately 50g will be processed for moisture content and organic matter after removing roots, drying at 105°C for 24hr, weighing, combusting at 450°C for 4hr and reweighing. The remainder of the sample will be processed for root mass, stone and C content by removing, sorting and dry air-drying soils for 4d or until constant weight. We will use a Retsch mill to homogenize a 50g subsample and submit to the Center for C and N content using a Thermo Flash EA. For total N, we will homogenize a 50g subsample in a Wiley mill. Sample homogenization will conform to EPA standards for soil C. We will include replicate processes to quantify the precision of laboratory methods.

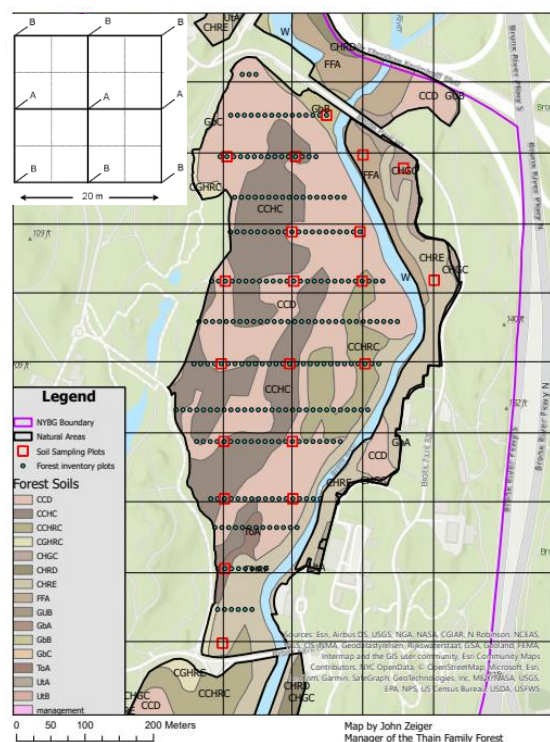


Figure 2: Sampling strategy relative to existing CFI plots, soil types and regular 1 ha grid. Inset plot map shows central transect “A” in 48 CFI plots with individual litter and soil samples to 0-10, 10-20, 20-50 cm depth. Deeper samples in “A”, litter and the top three depths in “B” are aggregated at the plot x depth level.

mass, stone and C content by removing, sorting and drying roots at 60°C for 3d, weighing them and then air-drying soils for 4d or until constant weight. We will sieve dried soil to 2 mm, weigh stones, and use a Retsch mill to homogenize a 50g subsample and submit 2g to the CUNY Advanced Science Research Center for C and N content using a Thermo Flash EA. For litter samples, we will dry at 60°C for 3d and homogenize a 50g subsample in a Wiley mill. Sample holding times and chain of custody requirements will conform to EPA standards for soil C. We will include 5% technical replicates of any subsampling processes to quantify the precision of laboratory methods. We will archive up to 50ml of remaining air-

dried soil samples in conical tubes for long term storage in the NYBG laboratory collection. Data will be quality checked and entered in the TFF database.

Objective 2: We will deploy a weather station (e.g. Onset U30) for continuous monitoring of rainfall, air temperature, relative humidity, soil temperature and moisture content in two locations in adjacent CFI plots closest to existing sap flux and stream gauge monitors. We will download data quarterly, quality check for sensor and power error and upload data to the TFF database.

Objective 3: We will collect a 100g subsample of air-dried soil and submit them for basic soil health (pH, and available P, K, Ca, Mg, S, micronutrients, reactive Al, CEC, base saturation, organic matter) and heavy metal testing (Pb, Cr, Cd, Ni) at the UVM agricultural and environmental testing lab. Data will be quality checked and entered to the TFF database.

Objective 4: The field data and sampling will be supported by the NYBG internship program which provides 10-20 local high-school and undergraduate students with immersive experiences across NYBG departments. PI Oberle and Co-PI Zeiger will follow best practices for safe and inclusive fieldwork and provide opportunities for interns and the lab technician to interact with peers and present their work as part of the summer intern symposium. The project activities, from SOP development to sample archiving and database hosting, will become a routine addition to long-term monitoring through the TFF CFI program pending funds. PI Oberle will present results to Forests For All New York City (FFANYC).

Outcomes and Timeline: Activities associated with each objective will generate concrete outcomes for key stakeholders (Table 1), including the first soil C map of the TFF, the first continuous soil climate record, a comparison of long-term trends in soil health focusing on heavy metals and immersive training for interns and technician. The project outcomes will immediately benefit diverse stakeholders through PI's role in FFANYC. This monitoring will inform the program of adaptive management to improve the ecological integrity of the TFF (Schuler and Forrest (2016).

Activity (Objective)	Su'24	Fa'24	Wi'25	Sp'25	Stakeholders	Outcome
Hiring & training (4)	x				NYBG, NYC	local youth participate
Field sampling (all)	x				NYBG, VT-LTSM, SI ForestGEO	SOPs, QA objectives developed
Weather Station deployment (3)	x				NYBG, USGS, SCAN	9 mo of quality soil microclimate
Lab SBD & C (1)		x			NYBG, VT-LTSM, SI ForestGEO	1st quality C map of TFF
Lab Soil Health & Metals (2)		x			NYBG, VT-LTSM, USACE	1st long-long term heavy metal change
Analysis (all)			x		NYBG, FEMC, FFANYC	FEMC meeting
Datasharing (all)				x	NYBG, FFANYC, FEMC	TFF database to FEMC standard
Final Reporting (all)				x	NYBG, FEMC	Final reporting complete

Project Roles: PI Oberle will develop Standard Operating Procedures, including QA processes, train interns, supervise the lab technician, analyze data, participate in the 2025 FEMC Annual Conference. Co-PI Zeiger (M.S. Manager, TFF) will supervise interns and contribute to data analysis.

Resources: PI Brad Oberle (Ph.D. Associate Curator) has more than 15 years of experience designing, collecting and analyzing forest inventory and ecosystem data, including successful completion of an EPA Cooperative Agreement MX 00D680-18/6207, which included forest C and continuous water monitoring (Oberle et al. 2024) as well as certification in Good Laboratory Practices by the CITI Program.

Co-PI John Zeiger (M.S. Manager, TFF) has 10 years of conservation and land management experience.

Pfizer Plant Research Laboratory The 23,000 ft² research lab contains all equipment and supplies necessary for processing ecological samples, including 2 drying ovens, a muffle furnace, an array of balances with different levels of precision, a Wiley Mill 4 and a Retsch bead mill.

IT NYBG is a global leader in botanical databasing, digitization and public data access. It has a 650-user institutional LAN supporting a variety of operating systems including Windows, MacOS, and Linux. It includes over 50 servers for departmental applications and files, the on-line library catalog, image storage, the Virtual Herbarium, backup facilities, email, and various NYBG web sites including a new initiative to provide a permanent, secure, publicly available ecological data repository for the TFF.

Literature Cited

- Britton, N.L. 1906. "The Hemlock Grove on the Banks of the Bronx River and What It Signifies." *Contributions from the New York Botanical Garden* 4 (83): 5–13.
<https://www.biodiversitylibrary.org/bibliography/4978>
- McDonnell, M. J., Pickett, S. T., Groffman, P., Bohlen, P., Pouyat, R. V., Zipperer, W. C., ... & Medley, K. (1997). Ecosystem processes along an urban-to-rural gradient. *Urban ecosystems*, 1, 21-36.
- Nagele, E., Ashton, M. S., Harris, T. B., Forrest, T. A., DeLuca, E. A., Martello, M. E., & Duguid, M. C. (2024). A Century of Change in a Mature Urban Forest: the Thain Family Forest of the New York Botanical Garden, Bronx, New York. *Journal of Forestry*, fvad057.
- Nagele, E., Co, M., Forrest, T. A., DeLuca, E. A., Ashton, M. S., Harris, T. B., & Duguid, M. C. (2023). Using Geologic Features to Visualize Long-Term Changes in Urban Forest Structure and Composition: Thain Family Forest of the New York Botanical Garden, Bronx, New York. *Journal of Forestry*, fvad047.
- Oberle, B., Bressan, S., McWilliams, J., & Díaz-Almeyda, E. (2023). Urban food forestry transforms fine-scale soil function for rapid and uniform carbon sequestration. *Urban Ecosystems*, 26(5), 1239-1250.
- Oberle, B., Cole, P. O., Frank, G., Gates, A., Hall, B., Harvey, D., ... & Bustetter, S. P. (2023). Multilevel allometric growth equations improve accuracy of carbon monitoring during forest restoration. *Trees, Forests and People*, 14, 100442.
- Pouyat, R. V., & McDonnell, M. J. (1991). Heavy metal accumulations in forest soils along an urban-rural gradient in southeastern New York, USA. *Water, Air, and Soil Pollution*, 57, 797-807.
- Pregitzer, C. C., Hanna, C., Charlop-Powers, S., & Bradford, M. A. (2022). Estimating carbon storage in urban forests of New York City. *Urban Ecosystems*, 25(2), 617-631.
- Price, R. (1976). *History of Forest Service research in the central and southern Rocky Mountain regions: 1908-1975* (Vol. 27). Rocky Mountain Forest and Range Experiment Station, Forest Service, US Department of Agriculture.
- Schuler, J.A. and T.A. Forrest. 2016. *NYBG Thain Family Forest Program: 2008-2025*. Bronx, NY: The New York Botanical Garden.
- Smithsonian Tropical Research Institute "Protocol for Soil Carbon Sampling in ForestGEO Census Plots". (n.d.). Retrieved March 20, 2024, from https://forestgeo.si.edu/sites/default/files/soil_carbon_sampling_protocol.pdf
- Ward, E. B., Doroski, D. A., Felson, A. J., Hallett, R. A., Oldfield, E. E., Kuebbing, S. E., & Bradford, M. A. (2021). Positive long-term impacts of restoration on soils in an experimental urban forest. *Ecological Applications*, 31(5), e02336.
- White, C. S., & McDonnell, M. J. (1988). Nitrogen cycling processes and soil characteristics in an urban versus rural forest. *Biogeochemistry*, 5, 243-262.

FEMC ECOSYSTEM MONITORING FUND 2024 RFP

NEW YORK BOTANICAL GARDEN

BUDGET JUSTIFICATION

FEMC FUNDS TOTAL COST (\$19,952)

A. INDIVIDUAL PERSONNEL (\$10,007)

A1. PI Oberle (\$1,827)

PI Oberle will develop Standard Operating Procedures for lab work, train interns, supervise the lab technician, analyze data and participate in the 2025 FEMC Annual Conference. Salary is calculated from \$95,000 base 1 wk FTE.

A2. Co-PI Zeiger (\$1,680)

Co-PI Zeiger will supervise interns for field sampling and contribute to data analysis. Salary is calculated from \$42 / hr, 40 hrs / wk, 1 week FTE.

A2. Lab Technician (\$6,500)

The part-time Lab Technician will perform laboratory methods necessary to estimate soil carbon stocks (e.g. homogenization, subsampling, root processing, moisture content, dry mass) and metal concentrations in surface soils and litter (air drying, homogenization, subsampling). Wages are calculated as \$20 / hr, 25 hr / wk, 13 wk appointment.

B. PERSONNEL FRINGE BENEFITS (\$2,322)

A1. PI Oberle (\$731)

Fringe for full-time salaried employees is calculated as 40% of wages (A1 = \$1,827)

A2. Co-PI Zeiger (\$941)

Fringe is calculated at \$23.52 per hour, 40 hrs / wk, 1 wk FTE

A3. Lab Technician (\$650)

Fringe for part-time employees is calculated as 10% of wages (A3 = \$7,000)

C. TRAVEL (\$860)

C1. FEMS Conference Travel (\$860)

Travel budget will allow PI Oberle to participate in the 2025 FEMC Annual Conference in Burlington, VT. Costs are estimated based on 2023 registration fee (\$60), two nights hotel accommodation (\$150 / night), three days for rental car (\$90 / day), \$80 fuel, and three days per diem food (\$50 / day).

D. SUPPLIES (\$4,950)

D1. Analytical Fees (\$2,566)

Funds are requested for fees to quantify soil chemistry in the long-term nested sampling design from the UVT Agricultural and Environmental Testing Lab. For all 20 20x20 m sampling plots, we will analyze physically aggregated surface samples (0-10 cm) for the basic test (P, K, Ca, Mg, S, micronutrients, reactive Al, CEC, base saturation, organic matter; \$17 / sample) and for heavy metals (lead, chromium, cadmium, nickel; \$10 / sample) as well as litter samples for heavy metals (\$10 / sample). For all 48 included long-term monitoring plots, we will measure the same variables at the same depths using the same lab (\$37 per sample total). Total cost includes \$50 for shipping.

D2. Weather Station (\$2,234)

Funds are requested for a weather station to continuously monitor microclimate and soil moisture at two depths in two locations near adjacent long-term inventory plots. Costs are estimated from the current list price of an Onset U30 USB weather station with solar power, shielded air temperature and relative humidity, rainfall gauge, 4 soil moisture probes, 2 soil temperature probes, analysis software, and shipping at current list prices.

D3. Data hosting (\$150)

Funds are requested to contribute to maintenance of the NYBG Digital Collections, which will host the database consistent with the Data Management Plan and FEMC requirements.

E. UTILITIES (NO FUNDS REQUESTED)

F. INDIRECT COST (\$1,814)

Indirect costs are calculated as the maximum allowable (10%) of the modified total direct cost of \$18,038.

NON-FEDERAL MATCH TOTAL (\$19,952)

A. INDIVIDUAL PERSONNEL (\$5,334)

A1. PI Oberle (\$3,654)

PI Oberle will contribute two weeks FTE in-kind effort. Salary is calculated from \$95,000 base 2 wk FTE.

A2. Co-PI Zeiger (\$1,680)

Co-PI Zeiger will contribute one week FTE in-kind effort. Salary is calculated from \$42 / hr, 40 hrs / wk, 1 week FTE.

B. PERSONNEL FRINGE BENEFITS (\$2,402)

B1. PI Oberle (\$1,462)

Fringe for full-time salaried employees is calculated as 40% of wages (A1 = \$3,654).

A2. Co-PI Zeiger (\$941)

Fringe is calculated at \$23.52 per hour, 40 hrs / wk, 1 wk FTE

C. TRAVEL (NO MATCH PROVIDED)

D. SUPPLIES (\$4,363)

D1. Analytical fees (\$2,920)

The Oberle Lab at NYBG will pay fees to quantify C and N content in the long-term nested sampling design. For all 20 20x20 m sampling plots, we will analyze physically aggregated surface samples at 4 standardized depths (0-10cm, 10-20 cm, 20-50cm, 50-100cm) and that of litter at the Core Facility at the City University of New York (estimate \$10 / sample for 5 samples / plot and 20 plots). For all 48 included long-term monitoring plots, we will measure C and N content at the three top depths (0-10cm, 10-20 cm, 20-50cm) and in the litter layer. PI Oberle has funds budgeted in his start-up (non-federal) for this purpose.

D4. Field (\$1,444)

Field supplies will include sample bags, packaging, field notebooks and measurement tools that will be purchased by the Oberle Lab using his start-up.

E. UTILITIES (NO MATCH PROVIDED)

F. INDIRECT COST (\$7,852)

Indirect cost match represents the unrecovered portion of the NYBG Federally Negotiated Rate of 53.29% of indirect costs of the project MTDC of \$18,038 less the allowable maximum of 10%.

FEMC ECOSYSTEM MONITORING FUND 2024 RFP**NEW YORK BOTANICAL GARDEN****BUDGET DETAIL**

Item	FEMC Funds	Non-Federal Matching	Total
A. Individual personnel			
A1. PI Oberle	\$ 1,827	\$ 3,654	\$ 5,481
A2. Co-PI Zeiger	\$ 1,680	\$ 1,680	\$ 3,360
A3. Lab Technician	\$ 6,500	\$ -	\$ 6,500
A. SUBTOTAL	\$ 10,007	\$ 5,334	\$ 15,341
B. Personnel fringe benefits			
B1. PI Oberle	\$ 731	\$ 1,462	\$ 2,192
B2. Co-PI Zeiger	\$ 941	\$ 941	\$ 1,882
B3. Lab Technician	\$ 650	\$ -	\$ 650
B. SUBTOTAL	\$ 2,322	\$ 2,402	\$ 4,724
C. Travel			
C1. FEMC Conference 2025	\$ 860	\$ -	\$ 860
C. SUBTOTAL	\$ 860	\$ -	\$ 860
D. Supplies			
D1. Analytical Fees	\$ 2,566	\$ 2,920	\$ 5,486
D2. Weather Station	\$ 2,234	\$ -	\$ 2,234
D3. Data hosting	\$ 150	\$ -	\$ 150
D4. Field and lab supplies	\$ -	\$ 1,444	\$ 1,444
D. SUBTOTAL	\$ 4,950	\$ 4,364	\$ 9,314
E. Utilities	\$ -	\$ -	\$ -
Modified Total Direct Cost	\$ 18,138	\$ 12,100	\$ 30,239
F. Indirect Costs	\$ 1,814	\$ 7,852	\$ 9,666
G. TOTAL COST	\$ 19,952	\$ 19,952	\$ 39,905

NYBG Data Management Plan

Modified from: Alexandra Kosiba and James Duncan, Forest Ecosystem Monitoring Cooperative

Data storage and organization

Data from this proposed project will be archived and organized into previously established cyberinfrastructure currently maintained and hosted by the New York Botanical Garden (NYBG, Bronx NY). Specifically, data collected by this project will populate The Thain Family Forest (TFF) database among NYBG Digital Collections (<https://sweetgum.nybg.org/science/digital-collections/>). The Digital Collections currently span 12 collection types, including the C. V. Starr Virtual Herbarium comprising approximately 4,500,000 herbarium specimens and approximately 3.6 million associated images (herbarium specimens and living plants). Digital collections are updated daily as NY pursues the goal of digitizing all of its plant and fungi specimens. The TFF database will meet FEMC standards including document archive, data display and visualization, search and download features, and metadata summaries (e.g. procedural methods, attribute information, related documents and citation and access specifications). The website will allow users to browse and search, and information about projects, datasets and associated publications will be indexed by search engines such as Google. Data will include all features necessary for tracing values to original records and verifying quality assurance criteria are met.

Metadata

The TFF database will follow an established metadata standard, similar to the Ecological Metadata Language Standard, for flexible documentation and sharing of metadata, and connecting to other digital collections.

NYBG Digital Collections, of which the TFF database will be a component, uses the Integrated Taxonomic Information System (<http://www.itis.gov/>) for documenting species information. This system, developed and maintained by a partnership of United States, Canadian and Mexican agencies, provides unique identifiers for species and maintains relationships between ‘accepted’ and ‘not accepted’ variants for that taxon. This allows identification of species taxa at the time of collection while permitting taxonomic definitions to change in the future without losing the original identification in a given dataset. Finally, the NYBG GIS Laboratory can support spatial features of the TFF Database.

Storage and Backup

NYBG Digital Collections file storage is provided by several NEC M110 disk arrays that currently occupy nearly 200TB that can scale to 300TB. Back-up of the data and systems is achieved using NEC’s Hydrastor which provides disc back-up with quicker speeds for back-up and restore and redundancy. We currently maintain two back-up systems, housed in different locations on the NYBG property, and powered by different power grids. The Virtual Herbarium website was completely revised in 2015, and in 2016, a new dual server configuration was installed that permits web access 24/7 even during the nightly backup and reindexing.

Access and Sharing

The data held in the repository will be available for exploration and download. Access constraints for data follow the Creative Commons (<https://creativecommons.org/>) licensing scheme. NYBG will make all data collected by this project publicly available with a permanent URL within one year of completing the project, consistent with FEMC requirements.